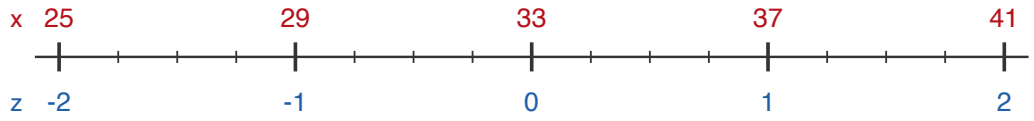
Answers: Z-Values

The mean is $\mu = 33$ and the standard deviation is $\sigma = 4$.

The z-value of a data value x is

$$z = \frac{x - \mu}{\sigma}$$

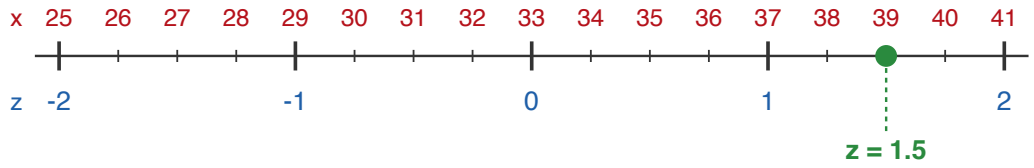
a) Fill in the x-values for each z-value on the number line.



$\mu = 33$ is at $z = 0$. Each step of one standard deviation adds or subtracts 4:

- $33 + 4 = 37$ and $37 + 4 = 41$
- $33 - 4 = 29$ and $29 - 4 = 25$

b) $x = 39$

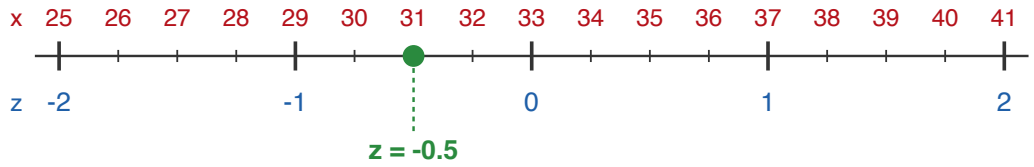


Visually: 39 is 6 above the mean. Each small tick is 1 and 4 ticks make one standard deviation, so 6 is $6/4 = 1.5$ standard deviations above the mean: $z = 1.5$.

Calculate:

$$z = \frac{39 - 33}{4} = \frac{6}{4} = 1.5$$

c) $x = 31$

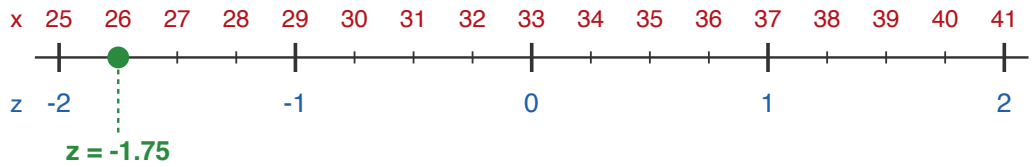


Visually: 31 is 2 below the mean. Each small tick is 1 and 4 ticks make one standard deviation, so 2 is $2/4 = 0.5$ standard deviations below the mean: $z = -0.5$.

Calculate:

$$z = \frac{31 - 33}{4} = \frac{-2}{4} = -0.5$$

d) $x = 26$

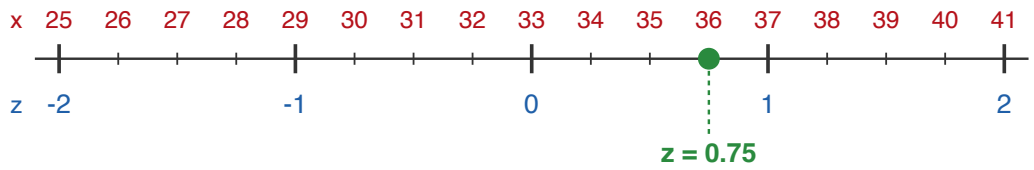


Visually: 26 is 7 below the mean. Each small tick is 1 and 4 ticks make one standard deviation, so 7 is $7/4 = 1.75$ standard deviations below the mean: $z = -1.75$.

Calculate:

$$z = \frac{26 - 33}{4} = \frac{-7}{4} = -1.75$$

e) $x = 36$



Visually: 36 is 3 above the mean. Each small tick is 1 and 4 ticks make one standard deviation, so 3 is $3/4 = 0.75$ standard deviations above the mean: $z = 0.75$.

Calculate:

$$z = \frac{36 - 33}{4} = \frac{3}{4} = 0.75$$